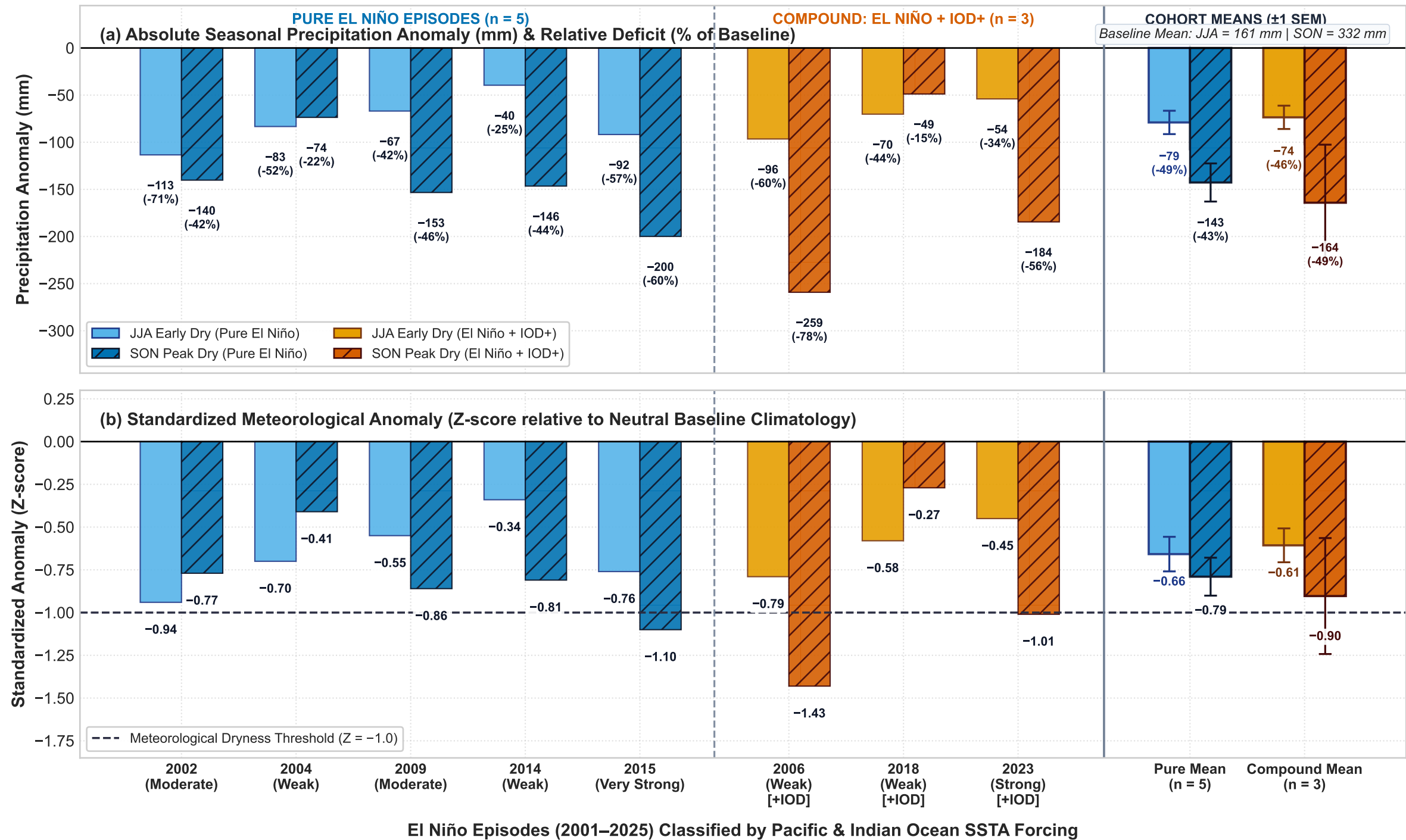


East Java Provincial Precipitation Response Across 25 Years of El Niño Episodes (2001–2025)



• Dataset: UCSB CHIRPS v2.0 (0.05° resolution resampled to 1 km) | Baseline Climatology: 6 neutral ENSO years (2001, 2003, 2012, 2013, 2019, 2025) within 2001–2025 multi-sensor domain
• Criteria: Pure ENSO based on NOAA CPC ONI ($\geq +0.5^{\circ}\text{C}$ for ≥ 5 consecutive seasons) | Compound IOD+ based on BoM/NOAA DMI ($\geq +0.4^{\circ}\text{C}$ in JJA/SON)
• Statistical Notation: Error bars on cohort composites indicate ± 1 standard error of the mean (SEM). Threshold line $Z = -1.0$ denotes meteorological dryness.